C1.3 Channel overview

This section introduces the signals that the ACE protocol provides, and where appropriate, describes their relationship to the existing AXI4 channels. The ACE protocol defines:

- signaling on existing AXI4 channels, see *Changes to existing AXI4 channels*
- signaling on ACE-specific channels, see *Additional channels defined by ACE*
- acknowledge signaling, see *Acknowledge signaling* on page C1-138.

Channel usage examples on page C1-139 gives examples of the use of the ACE signaling.

C1.3.1 Changes to existing AXI4 channels

Table C1-2 shows the ACE signals provided on existing AXI4 channels.

Table C1-2 ACE signals provided on existing AXI4 channels

AXI4 channel	Signal	Source	Description
Read address	**ARDOMAIN[1:0]**	Master	See *Read address channel (AR) signals* on page C2-152.
	ARSNOOP[3:0]	Master	
	ARBAR[1:0]	Master	
Write address	**AWDOMAIN[1:0]**	Master	See *Write address channel (AW) signals* on page C2-152.
	AWSNOOP[2:0]	Master	
	AWBAR[1:0]	Master	
Read data	**RRESP[3:2]**	Slave	See *Read data channel (R) signals* on page C2-152.

Note

There are no additional signals on the write data or write response channels.

C1.3.2 Additional channels defined by ACE

Three new channels are supported, these are the snoop address channel, the snoop data channel, and the snoop response channel.

The snoop address (AC) channel is an input to a cached master that provides the address and associated control information for snoop transactions.

The snoop response (CR) channel is an output channel from a cached master that provides a response to a snoop transaction. Every snoop transaction has a single response associated with it. The snoop response indicates if an associated data transfer on the CD channel is expected.

The snoop data (CD) channel is an optional output channel that passes snoop data out from a master. Typically, this occurs for a read or clean snoop transaction when the master being snooped has a copy of the data available to return.

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Table C1-3 shows the signals provided on the ACE-specific channels.

Table C1-3 ACE signals provided on ACE-specific channels

ACE-specific channel	Signal	Source	Description
Snoop address	**ACVALID**	Slave	See *Snoop address channel (AC) signals* on page C2-153.
	ACREADY	Master	
	ACADDR[ac-1:0][a]	Slave	
	ACSNOOP[3:0]	Slave	
	ACPROT[2:0]	Slave	
Snoop response	**CRVALID**	Master	See *Snoop response channel (CR) signals* on page C2-153.
	CRREADY	Slave	
	CRRESP[4:0]	Master	
Snoop data	**CDVALID**	Master	See *Snoop data channel (CD) signals* on page C2-154.
	CDREADY	Slave	
	CDDATA[cd-1:0][b]	Master	
	CDLAST	Master	

a. ac is the width of the snoop address bus.
b. cd is the width of the snoop data bus.

C1.3.3 Acknowledge signaling

ACE supports two additional acknowledge signals. These signals indicate that a master has completed a read or write transaction.

Table C1-4 shows the acknowledge signals.

Table C1-4 ACE read and write acknowledge signals

Signal	Source	Description
RACK	Master	See *Read acknowledge signal* on page C2-155
WACK	Master	See *Write acknowledge signal* on page C2-155

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C1.3.4 Channel usage examples

This section describes different examples of how the ACE channels are used when performing load and store operations. It describes:

- *Performing load operations from shareable locations*
- *Performing store operations to shareable locations*.

Performing load operations from shareable locations

The following procedure is an example of a master component loading data from a shareable address location, where the master component does not already have a copy of this location in its local cache:

1. The master component issues a read transaction on the read address channel.
2. The interconnect determines whether any other cache holds a copy of the location by passing the shareable address to other caching master components that can hold a copy, on the snoop address channel. In this context, these are snooped master components.
3. One of the following now occurs:
   - If any snooped master component holds the requested cache line, it is expected to indicate this by:
     - responding on the snoop response channel
     - providing the data to the interconnect, on the snoop data channel.

     The interconnect then provides the data, with an associated response, to the initiating master component on the read data channel.
   - If no snooped master component holds the requested cache line:
     - the interconnect initiates a transaction to main memory, effectively passing on the transaction from the initiating master component
     - the read data is supplied back to the master on the read data channel, as for standard transactions.
4. The master component indicates that the transaction has completed, using the **RACK** signal.

Note

If neither the initiating master nor the snooped cache take responsibility for writing a dirty cache line back to main memory at a later point in time, the interconnect might have to write data back to main memory at the same time that it is passed to the initiating master component.

If this occurs then the interconnect must generate the transaction address and write the dirty data returned from a snooped master component.

See *Transactions for performing load operations from shareable locations* on page C1-142 for more information.

Performing store operations to shareable locations

When a master stores to a cache line, to a shareable location, it removes all other copies of the cache line. This ensures that the master component has a unique copy of the cache line when it performs the store. The new value of the cache line at that location is propagated to other caches when respective caching master components subsequently read the cache line.

This section describes:

- *Store operations for a partial cache line* on page C1-140
- *Store operations for an entire cache line* on page C1-140
- *Store operations where the cache line is already cached* on page C1-141
- *Overlapping store operations* on page C1-141.

See *Transactions for performing store operations to shareable locations* on page C1-143 and *Transactions for accessing shareable locations when no cached copy is required* on page C1-143 for more information.

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Store operations for a partial cache line

A master component storing only a partial cache line must obtain a current copy of the cache line before performing the store. An example sequence is:

1. The initiating master component obtains a pre-store form of the cache line, and requests that other copies are removed, by issuing a ReadUnique transaction on the read address channel.
2. The interconnect passes the transaction to other caches on the snoop address channel.
3. Where applicable, a snooped master component responds to the transaction using the snoop response channel to indicate that it has the requested cache line. It also provides the cache line to the interconnect, using the snoop data channel.
4. The interconnect passes the cache line, together with a response, to the initiating master, using the read data channel.

   Note

   If no copies of the cache line are found during the snoop, a read of main memory is performed. The interconnect then passes the cache line and a response to the initiating master component, on the read data channel.

5. The master component performs the store and uses the **RACK** signal to indicate that the transaction has completed.

Note

While the cache line remains unique, loads and stores can be performed with no need for transactions to be broadcast to other caches.

Store operations for an entire cache line

A master component that is storing an entire cache line does not have to obtain data before storing the cache line. An example sequence is:

1. The initiating master component requests a unique copy of the cache line by issuing a MakeUnique transaction on the read address channel. This removes all other copies of the cache line.
2. The interconnect passes the transaction to other caches on the snoop address channel.
3. Snooped master components respond to the transaction using the snoop response channel to indicate that the cache line has been successfully removed.
4. A response is provided to the initiating master component, using the read data channel.

   Note

   Only the response fields are valid. No data transfer occurs.

5. The master component performs the store and uses the **RACK** signal to indicate that the transaction has completed.

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Store operations where the cache line is already cached

For a master component that already has a shared copy of the cache line, an example store sequence is:

1. The initiating master component requests a unique copy of the cache line by issuing a CleanUnique transaction on the read address channel. This removes all other copies of the cache line and writes any dirty copy to main memory.

   Note

   This transaction does not return the cache line to the initiating master component.

2. The interconnect passes the transaction to other caches on the snoop address channel. Snooped master components respond to the transaction using the snoop response channel to indicate:
   - that the cache line has been successfully removed
   - whether a dirty cache line must be written to main memory by the interconnect.
3. If a dirty cache line is being written to main memory, the appropriate snooped master provides the dirty cache line to the interconnect, using the snoop data channel. The interconnect then constructs the transaction to write the dirty cache line back to main memory.
4. A response is provided to the initiating master component, using the read data channel.

   Note

   Only the response fields are valid. No data transfer occurs.

5. The master component performs the store and uses the **RACK** signal to indicate that the transaction has completed.

Overlapping store operations

If two master components attempt simultaneous shareable store operations to the same cache line, the interconnect determines the order in which the transactions occur. This section uses the convention:

- master 1 is the component that the interconnect selects to proceed first
- master 2 is the component that the interconnect selects to proceed second.

Master 1 proceeds with the operation as described in *Performing store operations to shareable locations* on page C1-139.

Master 2 uses its snoop port to observe the master 1 store operation, and the following rules apply:

- If master 2 requires data, it receives the data when its own transaction completes, when it can proceed as normal with its own store operations.
- If master 2 is performing a full cache line store, it removes any original copy of the data when it observes the snoop transaction relating to the master 1 store. However, master 2 can then proceed with its own full cache line store when its own transaction completes.
- If master 2 is performing a partial line store, and originally had a copy of the cache line and therefore does not request a copy of the data then special consideration is required. In this case, when master 2 observes the snoop transaction relating to the master 1 store operation, it must remove its original copy of the data. Master 2 can then take one of the following options when its transaction completes:
  - For master 2 to retain the cache line in its cache, it must issue a new transaction to request a copy of the data, enabling it to complete the store operation.
  - Master 2 can perform a partial line write to main memory, ensuring the line is updated correctly, but the master does not retain a copy of the cache line in its cache. To access the cache line at a later point in time it must fetch the data again.

See *Sequencing transactions* on page C6-227 for more information.

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